

Sequences and series - Set 1 - Solutions

Specification MM2.1, MM2.2, MM2.3, M4.19.

Answer key

Question	Answer	Topic
1	B	Sequences and series
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Worked solutions

Question 1 SEQ-01 · Sequences and series · spec MM2.2

An arithmetic series has first term 5 and common difference 3. Find the sum of the first 40 terms.

Answer: B

$$S_{40} = \frac{40}{2}(2 \cdot 5 + 39 \cdot 3) = 20(10 + 117) = 2540.$$

Full method - how and why

A direct application of the arithmetic sum formula. The only decision is which version of the formula to use.

- 1 Identify $a = 5$, $d = 3$, $n = 40$.

why: 'First term' and 'common difference' name a and d exactly; n is how many terms you are adding.

- 2 Use $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{40} = 20[10 + 39 \times 3]$.

why: Use this version when you know a and d ; the other version $\frac{n}{2}(a + l)$ needs the last term first.

- 3 Evaluate the bracket: $10 + 117 = 127$, so $S_{40} = 20 \times 127 = 2540$.

why: Note $(n-1)d = 39 \times 3$, not 40×3 - there are 39 steps between 40 terms.

Where it goes wrong

- Using $40d$ instead of $39d$, giving 2600.
- Forgetting to halve n , or halving the bracket instead.

Takeaway $S_n = \frac{n}{2}[2a + (n-1)d]$ - the number of **gaps** is $n-1$, one fewer than the number of terms.

Question 2 SEQ-02 · Sequences and series · spec MM2.3

Find the sum to infinity of the geometric series $12 - 4 + \frac{4}{3} - \frac{4}{9} + \dots$

Answer: C

$$a = 12, r = -\frac{1}{3}. S_{\infty} = \frac{12}{1 + \frac{1}{3}} = \frac{12}{4/3} = 9.$$

Full method - how and why

A sum to infinity. Find the ratio (including its sign) and check $|r| < 1$ before using the formula.

- 1 Find r by dividing consecutive terms: $\frac{-4}{12} = -\frac{1}{3}$.

why: Always divide a term by the one before it; the alternating signs mean r is negative.

- 2 Check convergence: $|- \frac{1}{3}| < 1$ ✓.

why: The sum to infinity only exists when $|r| < 1$; otherwise the formula is meaningless.

- 3 Apply $S_{\infty} = \frac{a}{1-r} = \frac{12}{1 - (-\frac{1}{3})} = \frac{12}{\frac{4}{3}}$.

why: $1-r$ with a negative r becomes $1 + \frac{1}{3}$ - the double negative is the main trap.

- 4 Divide: $12 \times \frac{3}{4} = 9$.

why: Dividing by a fraction is multiplying by its reciprocal.

Where it goes wrong

- Using $r = +\frac{1}{3}$ and getting 18.
- Computing $\frac{12}{\frac{4}{3}}$ as 16 by multiplying instead of dividing.

Takeaway $S_{\infty} = \frac{a}{1-r}$; if r is negative the denominator grows, so the sum is smaller than $a/(1-|r|)$.

Question 3 SEQ-03 · Sequences and series · spec MM2.1

A sequence is defined by $x_{n+1} = 3x_n - 2$ with $x_1 = 2$. Find x_5 .

Answer: D

$$x_2 = 4, x_3 = 10, x_4 = 28, x_5 = 82.$$

Full method - how and why

A recurrence with a small index - just iterate. There is no shortcut worth finding for only four steps.

1 Apply the rule repeatedly from $x_1 = 2$: $x_2 = 3(2) - 2 = 4$.

why: $x_{n+1} = 3x_n - 2$ means 'triple the last term, subtract 2'.

2 $x_3 = 3(4) - 2 = 10$; $x_4 = 3(10) - 2 = 28$; $x_5 = 3(28) - 2 = 82$.

why: Write each step on its own line - the numbers grow quickly and a single slip propagates.

Where it goes wrong

- Stopping at $x_4 = 28$ (an offered option) by miscounting the steps: going from x_1 to x_5 takes **four** applications.
- Reading the rule as $3(x_n - 2)$, which gives $x_5 = 162$ - also offered.

Takeaway Count the applications carefully: reaching x_k from x_1 takes $k - 1$ steps.

Question 4 SEQ-04 · Sequences and series · spec MM2.2

The n th term of an arithmetic sequence is $u_n = 7 - 4n$. Find the sum of the first 30 terms.

Answer: A

$$u_1 = 3, u_{30} = 7 - 120 = -113. S_{30} = \frac{30}{2}(3 - 113) = 15(-110) = -1650.$$

Full method - how and why

You are given the n th term rather than a and d . Generate the first and last terms and use the $\frac{n}{2}(a + l)$ form.

1 First term: $u_1 = 7 - 4 = 3$. Common difference: the coefficient of n is -4 , so $d = -4$.

why: In $u_n = An + B$ the gradient A is the common difference - the sequence is linear in n .

2 Last term: $u_{30} = 7 - 120 = -113$.

why: Substitute $n = 30$ directly.

3 Use $S_n = \frac{n}{2}(a + l) = 15(3 + (-113)) = 15 \times (-110)$.

why: This version is quickest when the first and last terms are easy to compute.

4 Evaluate: -1650 .

why: Negative, as expected - most terms are negative.

Where it goes wrong

- Taking $d = 7$ or $a = 7$ from the formula rather than computing u_1 .
- Sign slips when adding $3 + (-113)$.

Takeaway For $u_n = An + B$: $d = A$ and $u_1 = A + B$; then either sum formula works.

Question 5 SEQ-06 · Sequences and series · spec MM2.2

The sum of the first n terms of a series is $S_n = 2n^2 + 3n$. Find the 10th term.

Answer: F

$$u_{10} = S_{10} - S_9 = (200 + 30) - (162 + 27) = 230 - 189 = 41.$$

Full method - how and why

Given a formula for S_n , individual terms come from the difference of consecutive sums.

- 1 Use $u_{10} = S_{10} - S_9$.
why: The 10th term is what the total gains when you add it - this works for any series, arithmetic or not.
- 2 $S_{10} = 2(100) + 30 = 230$.
why: Substitute $n = 10$.
- 3 $S_9 = 2(81) + 27 = 189$.
why: Substitute $n = 9$; $2 \times 81 = 162$.
- 4 Subtract: $230 - 189 = 41$.
why: Both wrong intermediate values (230 and 189) are offered as distractors.

Where it goes wrong

- Answering 230 (that is the sum of ten terms, not the tenth term).
- Substituting $n = 10$ into a 'term' formula that was never given.

Takeaway $u_n = S_n - S_{n-1}$ recovers terms from a sum formula.

Question 6 SEQ-08 · Sequences and series · spec MM2.3

The second term of a geometric sequence is 6 and the fifth term is 48. Find the sum of the first 6 terms.

Answer: F

$$ar = 6, ar^4 = 48, \text{ so } r^3 = 8, r = 2, a = 3. S_6 = 3 \cdot \frac{2^6 - 1}{2 - 1} = 3 \times 63 = 189.$$

Full method - how and why

Two terms of a geometric sequence determine a and r . Dividing the two equations isolates r .

- 1 Write the facts: $ar = 6$ and $ar^4 = 48$.
why: The k th term is ar^{k-1} ; the second term has r^1 , the fifth has r^4 .
- 2 Divide: $\frac{ar^4}{ar} = r^3 = \frac{48}{6} = 8$, so $r = 2$.
why: Division cancels a - always the first move with two geometric terms.
- 3 Back-substitute: $a(2) = 6$, so $a = 3$.
why: Use the simpler of the two equations.
- 4 Sum: $S_6 = a \frac{r^6 - 1}{r - 1} = 3 \times \frac{64 - 1}{1} = 189$.
why: With $r = 2$ the denominator is 1, making the arithmetic easy.

Where it goes wrong

- Using $r^4/r^1 = r^4$ instead of r^3 and getting a non-integer ratio.
- Computing S_6 with $2^6 = 32$ instead of 64.

Takeaway Divide two geometric terms to find r ; the index difference tells you which power of r you get.

Question 7 SEQ-09 · Sequences and series · spec MM2.2

How many terms of the arithmetic series $3 + 7 + 11 + \dots$ are needed for the sum to first exceed 500?

Answer: B

$$S_n = \frac{n}{2}(6 + 4(n - 1)) = n(2n + 1). \quad n = 15: 465; \quad n = 16: 528 > 500. \text{ So } 16.$$

Full method - how and why

An inequality on a sum. Form S_n , then find the first integer n that works - by trial, since n must be a whole number.

- 1 Identify $a = 3$, $d = 4$ and form the sum: $S_n = \frac{n}{2}[6 + 4(n - 1)] = \frac{n}{2}(4n + 2) = n(2n + 1)$.

why: Simplifying to $n(2n + 1)$ makes the trial values much easier to compute.

- 2 Test $n = 15$: $15 \times 31 = 465$, which is less than 500.

why: Estimate first: $2n^2 \approx 500$ gives $n \approx 15.8$, so start near there.

- 3 Test $n = 16$: $16 \times 33 = 528 > 500$ ✓.

why: The sum is increasing, so the first n that exceeds 500 is the answer.

- 4 Answer: 16 terms.

why: Check the one below to be sure it fails - that is what 'first exceeds' demands.

Where it goes wrong

- Solving $2n^2 + n = 500$ and rounding down to 15.
- Forgetting that n must be a positive integer and quoting a decimal.

Takeaway For 'first exceeds' questions, solve approximately then test the two neighbouring integers.

Question 8 SEQ-10 · Sequences and series · spec MM2.3

A geometric series has first term 2 and common ratio $r > 0$. The sum of the first three terms is 14. Find the sum of the first five terms.

Answer: F

$$2 + 2r + 2r^2 = 14 \Rightarrow r^2 + r - 6 = 0 \Rightarrow r = 2. \quad S_5 = 2(2^5 - 1) = 62.$$

Full method - how and why

The first three terms give a quadratic in r ; the positivity condition picks one root.

- 1 Write the sum of three terms: $2 + 2r + 2r^2 = 14$.

why: Terms are a , ar , ar^2 with $a = 2$.

- 2 Divide by 2 and rearrange: $r^2 + r - 6 = 0$.

why: Dividing first keeps the numbers small.

- 3 Factorise: $(r + 3)(r - 2) = 0$, so $r = 2$ or $r = -3$. The condition $r > 0$ gives $r = 2$.

why: The question's restriction exists precisely to discard the negative root.

- 4 Sum five terms: $S_5 = 2 \frac{2^5 - 1}{2 - 1} = 2 \times 31 = 62$.

why: $2^5 = 32$, so the bracket is 31.

Where it goes wrong

- Using $r = -3$ and getting $S_5 = 2 \times \frac{-243 - 1}{-4} = 122$.
- Adding two more terms by hand and slipping: $2 + 4 + 8 + 16 + 32 = 62$ is a good check.

Takeaway Conditions like ' $r > 0$ ' are there to select between roots - apply them before continuing.

Question 9 SEQ-12 · Sequences and series · spec MM2.3

Which of the following geometric series is/are convergent?

I $1 + 0.9 + 0.81 + \dots$

II $3 - 3 + 3 - 3 + \dots$

III $100 + 10 + 1 + 0.1 + \dots$

Answer: F

A geometric series converges if and only if $|r| < 1$. I has $r = 0.9$, III has $r = 0.1$: both converge. II has $r = -1$ and does not converge.

Full method - how and why

Convergence of a geometric series depends only on $|r| < 1$. Compute r for each and apply the test.

1 I: $r = \frac{0.9}{1} = 0.9$, and $|0.9| < 1$: **converges**.

why: A ratio just below 1 still converges, though slowly.

2 II: $r = \frac{-3}{3} = -1$, and $|-1| = 1$ is not less than 1: **does not converge**.

why: The partial sums oscillate between 3 and 0 forever - they never settle, so there is no sum to infinity.

3 III: $r = \frac{10}{100} = 0.1$: **converges**.

why: Small ratio, rapid convergence.

4 I and III only.

why: Match the verdicts to the option list.

Where it goes wrong

- Thinking II converges to 0 or to 1.5 because the terms 'average out' - the partial sums must approach a single limit.
- Judging by the size of the terms rather than the ratio: III starts at 100 but converges fastest.

Takeaway Convergence is decided by $|r| < 1$ alone - not by the size of the first term or by whether the terms alternate.

Question 10 SEQ-13 · Sequences and series · spec MM2.2

An arithmetic sequence has $u_3 = 10$ and $u_8 = 25$. Find the smallest value of n for which $u_n > 100$.

Answer: C

$$5d = 15 \Rightarrow d = 3, a = 4, u_n = 4 + 3(n - 1) = 3n + 1 > 100 \Rightarrow n > 33, \text{ so } n = 34.$$

Full method - how and why

Two terms give a and d ; then solve an inequality for n and round to the correct integer.

1 Subtract the two facts: $u_8 - u_3 = 5d = 25 - 10 = 15$, so $d = 3$.

why: From u_3 to u_8 is five steps, so the difference is $5d$ - count the gaps, not the terms.

2 Find a : $u_3 = a + 2d = 10$, so $a = 10 - 6 = 4$.

why: The third term is $a + 2d$, not $a + 3d$.

3 Write the general term: $u_n = 4 + 3(n - 1) = 3n + 1$.

why: Simplify before solving - it avoids bracket errors.

4 Solve $3n + 1 > 100 \Rightarrow n > 33$, so the smallest integer is $n = 34$.

why: Strict inequality: $n = 33$ gives exactly 100, which is not greater than 100.

Where it goes wrong

- Answering 33, where $u_{33} = 100$ exactly.
- Using $u_8 - u_3 = 8d - 3d$ correctly but then $u_3 = a + 3d$ incorrectly.

Takeaway u_p to u_q spans $q - p$ common differences; and 'greater than' excludes the boundary case.

Question 11 SEQ-14 · Sequences and series · spec MM2.1

The sequence $u_n = n^2 - 10n + 30$ is defined for positive integers n . What is the smallest value taken by any term of the sequence?

Answer: B

$$u_n = (n - 5)^2 + 5, \text{ minimised at } n = 5 \text{ with value } 5.$$

Full method - how and why

A quadratic sequence. Completing the square finds the minimum - but n must be a positive integer, so check that the optimum is attainable.

1 Complete the square: $n^2 - 10n + 30 = (n - 5)^2 + 5$.

why: Half of -10 is -5 ; $30 - 25 = 5$.

2 The square is smallest when $n = 5$, giving $u_5 = 5$.

why: $(n - 5)^2 \geq 0$ with equality only at $n = 5$.

3 $n = 5$ is a positive integer, so the minimum is attained: the smallest term is 5.

why: If the vertex had been at a non-integer (say 4.5), you would have to test the integers either side.

Where it goes wrong

- Giving $n = 5$ (the position) instead of 5 (the value) - here they coincide, which hides the error.
- Taking $u_1 = 21$ as the smallest by assuming sequences always increase.

Takeaway Minimise a quadratic sequence by completing the square, then check the nearest **integer** values of n .

Question 12 SEQ-15 · Sequences and series · spec MM2.3

Evaluate $\sum_{k=1}^{10} 3 \times 2^k$.

Answer: F

This is $6 + 12 + \dots + 6 \cdot 2^9$: first term 6, ratio 2, 10 terms: $6(2^{10} - 1) = 6 \times 1023 = 6138$.

Full method - how and why

Sigma notation hiding a geometric series. Write out the first term and the ratio, and count the terms carefully.

1 Write out the start: $k = 1$ gives $3 \times 2 = 6$; $k = 2$ gives 12. So $a = 6$, $r = 2$.

why: The first term is the value at the **lower limit**, not 3.

2 Count the terms: k runs from 1 to 10, which is 10 terms.

why: Upper minus lower plus one.

3 Apply the sum formula: $S_{10} = 6 \frac{2^{10} - 1}{2 - 1} = 6 \times 1023$.

why: $2^{10} = 1024$ is worth knowing by heart.

4 Evaluate: 6138.

why: $6 \times 1000 = 6000$ plus $6 \times 23 = 138$.

Where it goes wrong

- Taking $a = 3$ (ignoring the $k = 1$ factor of 2), giving 3069.
- Using 2^{10} instead of $2^{10} - 1$, giving 6144.

Takeaway In $\sum ar^k$ the first term is the value at the lower limit; count terms as (upper - lower + 1).